

ADISESHA

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PROFESSIONAL SUMMARY

Senior Data Scientist | Machine Learning Engineer with 8+ years of experience in data analytics, machine learning, predictive modeling, and data visualization. Adept at working with structured and unstructured data, leveraging advanced statistical analysis and machine learning algorithms to drive strategic business decisions. Proficient in the end-to-end data science lifecycle, including data acquisition, validation, exploratory data analysis (EDA), feature engineering, and dimensionality reduction (PCA). Skilled in developing and optimizing machine learning models such as decision trees, neural networks, random forests, SVMs, ensemble methods (boosting, bagging), and deep learning architectures (CNNs, LSTMs, transformers).

Expert in Natural Language Processing (NLP), applying text analytics, sentiment analysis, topic modeling, named entity recognition (NER), and speech recognition to extract insights from customer interactions, claims, and chat logs. Hands-on experience with SpaCy, NLTK, and deep learning-based NLP models for enhancing business intelligence and automation. Strong background in big data processing, with expertise in Hadoop, Spark, Hive, and PySpark for handling large-scale batch and streaming data. Experience in cloud-based ML solutions (AWS, Azure, GCP) for adaptable model deployment and infrastructure optimization. Advanced programming skills in Python, R, Scala, and SQL, with experience in relational databases (SQL Server, Oracle, Teradata) and NoSQL databases (MongoDB, Cassandra). Expertise in designing efficient queries, ETL pipelines, and real-time data processing to support business intelligence. Proficient in data visualization and reporting using Tableau, Power BI, and Looker, enabling stakeholders to make data-driven decisions through interactive dashboards and reports. Deep learning expertise includes working with TensorFlow and PyTorch to implement CNNs for image recognition, RNNs for time-series forecasting, and transformers for NLP applications. Experienced in MLOps and model deployment, integrating ML models into production environments using Docker, Kubernetes, and CI/CD pipelines. Skilled in migrating monolithic applications to microservices architecture, optimizing ML inference via REST APIs, and cloud-based deployment strategies. Proven track record of problem-solving, business acumen, and cross-functional collaboration, delivering actionable insights that drive operational efficiency, cost savings, and revenue growth. Known for effectively communicating complex data insights to technical and non-technical stakeholders, ensuring data-driven strategies align with business goals.

Technical Skills

- **Programming Languages & Frameworks:** Python, R, Scala, Java, C, C++, SQL, Shell
- **Frameworks:** Scikit-learn, TensorFlow, PyTorch, Keras, NLTK, SpaCy, XGBoost, Pandas, NumPy, SciPy, OpenCV, Flask, Docker
- **Data Science/Analytics Tools:** Jupyter Notebook, Anaconda, SAS, RStudio, MATLAB
- **Big Data Technologies:** Hadoop, HDFS, HIVE, Spark, PySpark, Sqoop
- **Version Control & Collaboration:** GitHub, Azure DevOps, Maven
- **Cloud Platforms & Services:** AWS (EC2, Lambda, Elastic Beanstalk, SageMaker, S3, DynamoDB, Athena, EMR, Lambda, API Gateway, CodePipeline), Azure (VMs, AKS, Azure ML, Blob Storage, Data Lake, HDInsight, Functions, ADF, DevOps), GCP (Compute Engine, GKE, Vertex AI, BigQuery, Cloud Functions, Cloud Run, Cloud Build), Oracle (Oracle Compute, OKE, Autonomous DB, MySQL, Big Data Service, Oracle Functions), IBM Cloud (Virtual Servers, Watson Studio, Cloud Pak for Data, Cloud Functions, DevOps)
- **Databases & Data Integration:** Relational: SQL Server, PostgreSQL, MySQL, Oracle, NoSQL: MongoDB, Cassandra, Data Integration: Kafka, Sqoop, AWS S3, Azure Storage
- **Machine Learning & Models:** Decision Trees, SVM, XGBoost, KNN, Neural Networks, Logistic Regression, Deep Learning: CNN, LSTMs, Transformers, NLP: NER, Sentiment Analysis, Text Analytics, Algorithms: PCA, ANOVA, Cross-Validation
- **Data Visualization & Reporting:** Tableau, Power BI, QlikView, Matplotlib, Seaborn, Plotly, MS Office, Crystal Reports, SSRS
- **Data Modeling & Design:** MS Visio, ERWIN, Star Schema, Snowflake, Normalization, Kimball & Inmon, Apache Kafka, AWS Glue
- **MLOps:** Flask, Docker, AWS Lambda, Azure ML, Vertex AI, Splunk, Grafana, AWS CloudWatch, Python, Statsmodels
- **Project Management & Collaboration:** Jira, GitHub, Azure DevOps, Sprint Planning, Scrum, Confluence, MS Teams, SharePoint

PROFESSIONAL EXPERIENCE

Fifth Third Bank

Senior Data Scientist/ML Engineer

Jan 2024 - Present

Evansville, IN

- **Quantitative Modeling Development & Management:** Developed the design and implementation of consumer credit risk quantitative models utilizing machine learning techniques, improving the accuracy of retail lending portfolio risk assessments. Successfully reduced credit risk prediction errors by 15%, enabling better-targeted risk mitigation strategies. Leveraged tools such as Scikit-learn, XGBoost, and Statsmodels for model development and optimization.

- **Advanced Analytics & Predictive Modeling:** Conducted advanced analytics by leveraging predictive modeling, machine learning, and optimization techniques to generate actionable insights. Developed models that resulted in a 20% improvement in predictive accuracy for customer creditworthiness and loan default rates, directly contributing to enhanced decision-making. Used Pandas, NumPy, Scikit-learn, and TensorFlow to build and refine models.
- **Model Development & Backtesting:** Maintained and developed 30+ machine learning models for consumer credit risk, continuously optimizing them based on feedback from business stakeholders. Increased model stability and reliability by 10%, enhancing risk management processes. Backtested models using SQL (PostgreSQL, MySQL) for data querying and Python for model validation and adjustments.
- **Ad-Hoc Reporting & Analytics:** Delivered actionable insights and reports for stakeholders through ad-hoc analysis requests, facilitating critical business decisions. Presented complex technical information to non-technical audiences, enabling clear understanding of model performance and business impact. Tools used include SQL Server, Tableau, and Power BI for report generation and data visualization.
- **Data Integration & Management:** Developed, integrated, and processed large and complex datasets, extracting valuable information from both structured and unstructured data sources. Reduced data preparation time by 30%, enabling faster decision-making. Utilized Hadoop, Spark, AWS S3, and Azure Storage for data storage, management, and integration of large datasets.
- **Collaboration with Stakeholders:** Collaborated with business stakeholders to understand needs and develop data-driven solutions. Played a pivotal role in gaining executive buy-in for machine learning-driven risk models, leading to the successful implementation of predictive models. Utilized GitHub and Azure DevOps for collaboration, version control, and project management.
- **Model Implementation & Experimentation:** Transformed business challenges into actionable plans for experimentation by developing data sources and applying relevant algorithms. Pioneered innovative experiments, improving model precision by 25% for identifying high-risk lending opportunities. Employed Flask, Docker, and AWS Lambda for model deployment and experimentation.
- **Cloud-based Model Deployment:** Utilized cloud services (AWS, Microsoft Azure, Google Cloud Platform) to deploy, manage, and monitor machine learning models. Successfully deployed over 10 AI/ML models to production environments, ensuring scalability and stability. Reduced deployment time by 40% by using AWS SageMaker, Azure ML, and Google Cloud's Vertex AI for streamlined cloud-based deployment.
- **Model Monitoring & Reporting Automation:** Developed automated systems for model monitoring, reporting, and performance evaluation. Reduced manual reporting time by 50%, improving operational efficiency. Integrated Splunk, Grafana, and AWS CloudWatch for real-time model monitoring and performance tracking.
- **Development of Reusable Codebase:** Contributed to the creation and maintenance of a reusable Python codebase for model development and analytics, accelerating the adoption of AI/ML solutions across the organization. This initiative decreased time-to-market by 35%.
- **Cross-functional team Collaboration:** Collaborated with data engineers, analysts, and business units to translate data science concepts into practical applications. Contributed to the development of AI solutions that increased operational efficiency by 15% in the healthcare industry. Collaborated using Azure DevOps for project tracking and team communication.
- **Applied machine learning algorithms and statistical models,** such as decision trees and NLP, using scikit-learn and MATLAB to identify data patterns, enhancing predictive accuracy.
- **Continuous Learning & Innovation:** Stayed and updated on AI/ML advancements, implementing cutting-edge techniques with SpaCy, NLTK, and XGBoost to enhance model performance by 20%.
- **Communication & Time Management:** Communicated technical findings, strategies, and recommendations to both technical and non-technical stakeholders. Managed multiple projects effectively, meeting deadlines and prioritizing tasks. Leveraged Microsoft Office (Word, Excel, PowerPoint) for clear, concise reporting and presentations.

Ascena Retail Group

Senior Data Scientist/ ML Engineer

May 2021 - Dec 2023

Patskala, Ohio

- Performed data mining, data collection, and cleansing to prepare datasets for machine learning models, automating processes using SQL, which improved data accuracy and efficiency in the RDBMS database.
- Created advanced machine learning algorithms, including decision trees and neural networks, which enhanced predictive accuracy and informed strategic decision-making.
- Delivered analytics use cases using Aster analytics functions, such as nPath and Text Parser, which optimized data insights and improved business strategies. Applied state-of-the-art ML/DL technologies to evaluate sentiment scores, improving model accuracy by re-training semantic segmentation models.
- Utilized classifiers developed in Scikit-learn with linguistic features from SpaCy, enhancing natural language processing capabilities and improving text analysis accuracy.
- Designed new language models (NLP/NLU) to understand and build predictions or summaries from customer calls, claim notes, webchats, and medical records, improving data analysis efficiency. Worked alongside data engineers to integrate models into a production system that can continuously be monitored and updated.
- Developed expertise in speech recognition, NLP, and NLU, applying these skills to enhance language processing capabilities. Designed a light-weight convolutional neural network (CNN) on TensorFlow to recognize keywords, phrases, tone, and inflection.
- Optimized machine learning models and automated processes through hyperparameter tuning, debugging, and troubleshooting, leading to improved model performance. Analyzed performance, and conducted experiments with different types of algorithms and models to identify the best algorithms to employ.
- Reported that CNN architecture in our application was leading to fast convergence and quick overfitting; Bi-direction LSTM architecture cost 10x training time, and it tended to be under fitting the training data initially.
- Migrated eDiscovery search engine from monolith to microservice architecture using NLTK, SciKit-Learn. Developed a DL module with Keras, using Glove (Wikipedia-trained word vectors) as the embedding layer.
- Conducted a hybrid of Hierarchical and K-means Cluster Analysis, which enhanced the accuracy of customer segmentation and informed targeted marketing strategies.

Molina healthcare

Apr 2019 - Apr 2021

Data Scientist

Bothell, WA

- Designed, and implemented machine learning models for patient risk stratification, readmission prediction, and disease progression forecasting, leveraging scikit-learn, TensorFlow, and PyTorch to improve healthcare outcomes.
- Extracted and analyzed EHR data from Cerner/Epic systems using SQL, Python (Pandas, NumPy), and Spark, providing actionable insights to enhance patient care and operational efficiency.
- collaboratively with senior leadership and cross-functional teams to translate complex analytical findings into actionable business insights, improving decision-making efficiency by 40%. Presented analytical findings and developed business cases to senior management, supporting the adoption of data-driven strategies and process improvements that optimized care models and drove cost-saving initiatives.
- Managed a team of 10+ data analysts and data scientists, ensuring the timely delivery of high-quality results on multiple analytical projects. Offered mentorship and technical guidance, which contributed to a 15% improvement in team productivity and project delivery. Fostered a culture of knowledge-sharing and continuous learning, implementing structured performance reviews and Spearheaded skill development initiatives, enhancing team retention by 20%.
- Developed the design and implementation of cloud-based data solutions using AWS S3, Oracle Autonomous Database, and Microsoft Azure, ensuring scalability, cost-efficiency, and seamless integration for large-scale data analytics. Streamlined data processing by leveraging big data technologies such as Hadoop and Apache Spark, boosting processing efficiency by 30% and enabling faster data insights for healthcare optimization.
- Utilized Oracle Data Integration, GoldenGate, and Apache Kafka to seamlessly integrate and move terabytes of data into cloud platforms, ensuring efficient data movement and integration. Collaborated with cross-functional teams to deploy cloud-based solutions that reduced system downtimes by 20% and improved operational performance.
- Monitored emerging policies within Medicaid, Medicare, and Marketplace programs using Oracle Cloud and Microsoft Azure, ensuring timely adaptations to new regulations. Developed and presented business cases and analytical frameworks that directly influenced management decision-making, improving provider payment accuracy by 15% and enabling enhanced care models.

Allvy Software Solutions

Jun 2017 - Jan 2019

Jr. Data scientist

Hyderabad, India

- Conducted two-factor, three-factor, and multi-factor ANOVA models, tested interactions, and derived random and mixed effect models, leading to improved understanding of data relationships.
- Performed comprehensive chi-square, t-test, Z-test, and non-parametric tests, leading to validated claims and conclusions based on p-values, enhancing decision-making accuracy.
- Fitted logistic regression, LDA, QDA, and KNN models for binomial classification, using Confusion Matrix and ROC Chart for evaluation, which improved model selection accuracy.
- Collaborated alongside data scientists, data engineers, and statisticians to analyze terabytes of data using Hadoop, SQL, Tableau, and Python. Developed a Python-based predictive model that combined physical modeling and telematics data to forecast the truck's loaded weight for maintenance prediction.
- Performed multiple linear regression using least squares and selection methods, recommending the best model, which enhanced predictive accuracy and model reliability. Constructed machine learning models using NumPy, SciPy, NLTK, SciKit Learn, MLPy, and OpenCV, improving prediction accuracy and efficiency.
- Automated reporting tasks to improve efficiency and reduce manual work using tools such as SQL, VBA, or Python. Tracked key performance indicators (KPIs) and other relevant metrics to monitor and optimize business operations. Analyzed performance data to help enhance business processes and decision-making. Contributed to maintaining and optimizing databases for smooth data extraction and reporting.
- Collaborated with Machine learning engineers to analyze the data based on their requirements. Experienced in creating pivot tables for analyzing data in Excel. Tested and ensured data accuracy through data integrity queries, debugging, and troubleshooting, enhancing data reliability and consistency.

KPIT Cummins Info Systems Ltd.

Jun 2016 - May 2017

Python Developer

Pune, India

- Developed an entire application from requirements gathering to deployment using Anaconda Python Environment, enhancing performance calculations with NumPy and SQL Alchemy.
- Created Python routines to log into websites and fetch data for selected options, improving data retrieval efficiency. Utilized Python modules such as urllib, urllib2, and Requests for web crawling, and applied ML techniques such as clustering, regression, and classification to enhance data analysis.
- Developed different Statistical Machine Learning and Data Mining solutions for business problems, generating data visualizations using R, Python, and Tableau to improve decision-making.
- Involved in the development of Web Services using SOAP for sending and getting data from the external interface in the XML format. Used with other packages such as Beautiful Soup for data parsing.
- Developed SQL queries and stored procedures on MySQL, enhancing database efficiency and performance. Analyzed and refactored code to reduce redundancy, improving code efficiency and maintainability. Designed and developed a text classification application using multiple models, enhancing text analysis accuracy and efficiency.
- Utilized Jira for defect tracking and project management, streamlining workflow, and enhancing team collaboration. Processed and analyzed data from CSV and Excel files, improving data accessibility for team projects.
- Participated in Sprint planning sessions and daily Agile SCRUM meetings, enhancing team collaboration and project alignment. Compiled detailed reports from application data, informing strategic decision-making and performance assessments.
- Maintained internal documentation and API specifications for team members. Stayed current with Python language advancements, libraries, and best practices to enhance software development processes. Actively participated in workshops, training, and knowledge-sharing sessions.